

FitBuddy AI

4. Project Planning Phase

Development schedule, milestones, responsibilities and risk planning.

Milestones

- M1 — Finalize project idea and requirements.
- M2 — Design application flow and UI.
- M3 — Implement FastAPI backend and routes.
- M4 — Integrate Gemini generation.
- M5 — Build Jinja2 pages and styling.
- M6 — Test inputs, API failures and generated output.
- M7 — Prepare documentation and demonstration.
- M8 — Final GitHub repository organization.

Suggested Work Breakdown

- Backend: FastAPI setup, routes, validation and response handling.
- AI integration: Gemini client, prompt construction and error handling.
- Frontend: Jinja2 templates, CSS and optional JavaScript.
- Testing: functional, validation, error and basic performance checks.
- Documentation: README, architecture, setup and demo material.

Risk and Mitigation

- API failure → show a user-friendly fallback/error message.
- Missing API key → validate configuration at startup or request time.
- Unexpected AI output → use structured prompts and post-processing/validation.
- Invalid user input → server-side validation.
- Accidental secret upload → use .gitignore and environment variables.

Completion Checklist

- Application starts successfully.
- Main user flow works end-to-end.
- AI generation works with the configured model/service.
- Templates render correctly.
- Errors are handled.
- Tests/checks are recorded.
- Documentation and demo evidence are ready.